

AYMANE AIT DADS

SDB ML Fusion Intern | PyTorch · Remote Sensing · Random Forest

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April 30, 2026

Hiring Team - MSc thesis assignment: Machine Learning Fusion of Sentinel 2 and ICESat-2 data for Satellite Derived Bathymetry (SDB)

Royal HaskoningDHV · Rotterdam, Netherlands

Dear Haskoning Hiring Team,

Your assignment calls for developing **machine learning models to predict bathymetry from multispectral inputs** - specifically fusing **Sentinel-2 imagery with ICESat-2 depth points** to produce scalable shallow-water depth estimates without costly in-situ surveys. At Orange Maroc, I built an end-to-end **ML pipeline** processing over **100,000 telemetry records** using **Random Forest** - one of the exact model architectures your literature review flags - delivering production-ready outputs directly used for business decisions.

Two proof points map directly to your technical scope. First, my **aerial image classification project** achieved **99.8% accuracy on 17,500+ overhead images** by benchmarking **CNN, DenseNet121, and CNN-Transformer hybrids** - the same multispectral spatial-pattern recognition your optical depth retrieval task demands. Second, I designed rigorous **ablation experiments** on model depth, learning rate schedules, and feature representations to quantify accuracy limits - exactly the kind of applicability and scalability analysis your assignment requires for engineering use cases.

Haskoning already holds recognized in-house SDB expertise and is now pushing toward a fully spaceborne solution - a methodological frontier where your team can turn this thesis into an immediately deployable tool. I want to contribute to that specific gap. I would welcome the chance to discuss how my background fits this project.

Sincerely,

AYMANE AIT DADS